Dev Board Introduction

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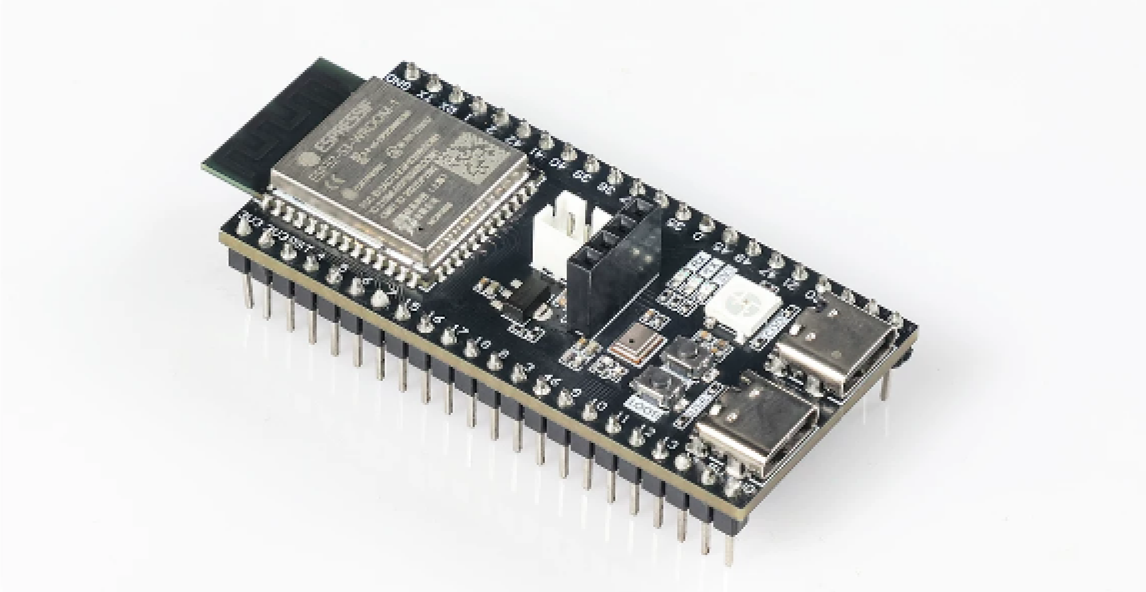
1. Development Board Overview

The core board used in this tutorial is the **ESP32-S3 full-featured development board**, equipped with the official Espressif module **ESP32-S3-WROOM-1-N16R8** (16 MB Flash + 8 MB Octal PSRAM). It features an on-board ES8311 audio codec + MEMS microphone + NS4150 amplifier + speaker connector + SSD1306 OLED display + WS2812 RGB LED, making it a full-featured teaching platform covering GPIO, audio, display, USB, and wireless communication.

1.1 Core Board Photo



1.2 Core Specifications at a Glance

Item	Specification
Main control module	ESP32-S3-WROOM-1-N16R8
Flash	16 MB (Octal SPI, N stands for Octal)
PSRAM	8 MB (Octal SPI)
PCB antenna	On-board ceramic / PCB antenna (built into WROOM-1)
USB-to-serial	CH340K (USB-C connector), auto-download circuit
Native USB OTG	USB-C connector (GPIO19 / GPIO20), supports Device / Host
Audio codec	ES8311 (I2S full duplex + I2C control)
Amplifier	NS4150 (3W Class D, enabled by GPIO7)
Microphone	On-board MEMS analog microphone (sampled through the ES8311 ADC)
Speaker connector	2-pin PH socket / solder pads
Display	SSD1306 128×32 monochrome OLED (I2C)
RGB LED	WS2812 × 1 (GPIO48, on-board LED)
Buttons	BOOT (GPIO0) + RST (EN)
Power supply	USB-C 5V → 3.3V LDO
Pin headers	2.54 mm standard spacing, dual-row, breadboard compatible
Dimensions	Approx. 55 × 28 mm

2. Module Details — ESP32-S3-WROOM-1-N16R8

2.1 Model Decoding

```
ESP32-S3-WROOM-1-N16R8
          |      || ||
          |      || |└── R8: 8 MB Octal PSRAM
          |      ||  └──── R: with PSRAM
          |      |└──────16: 16 MB Flash
          |      └──────── N: Octal SPI Flash
          └──────── WROOM-1: built-in PCB antenna module
```

Designator	Options	Meaning
Flash interface	`N` (Octal), `Q` (Quad)	N is faster and is the standard configuration for N16R8
Flash capacity	`4` / `8` / `16` (MB)	N16 = 16 MB
PSRAM capacity	`2` / `8` (MB)	R8 = 8 MB
Antenna	WROOM-1 (on-board) / WROOM-1U (IPEX external)	This board uses WROOM-1

2.2 Module Internal Structure

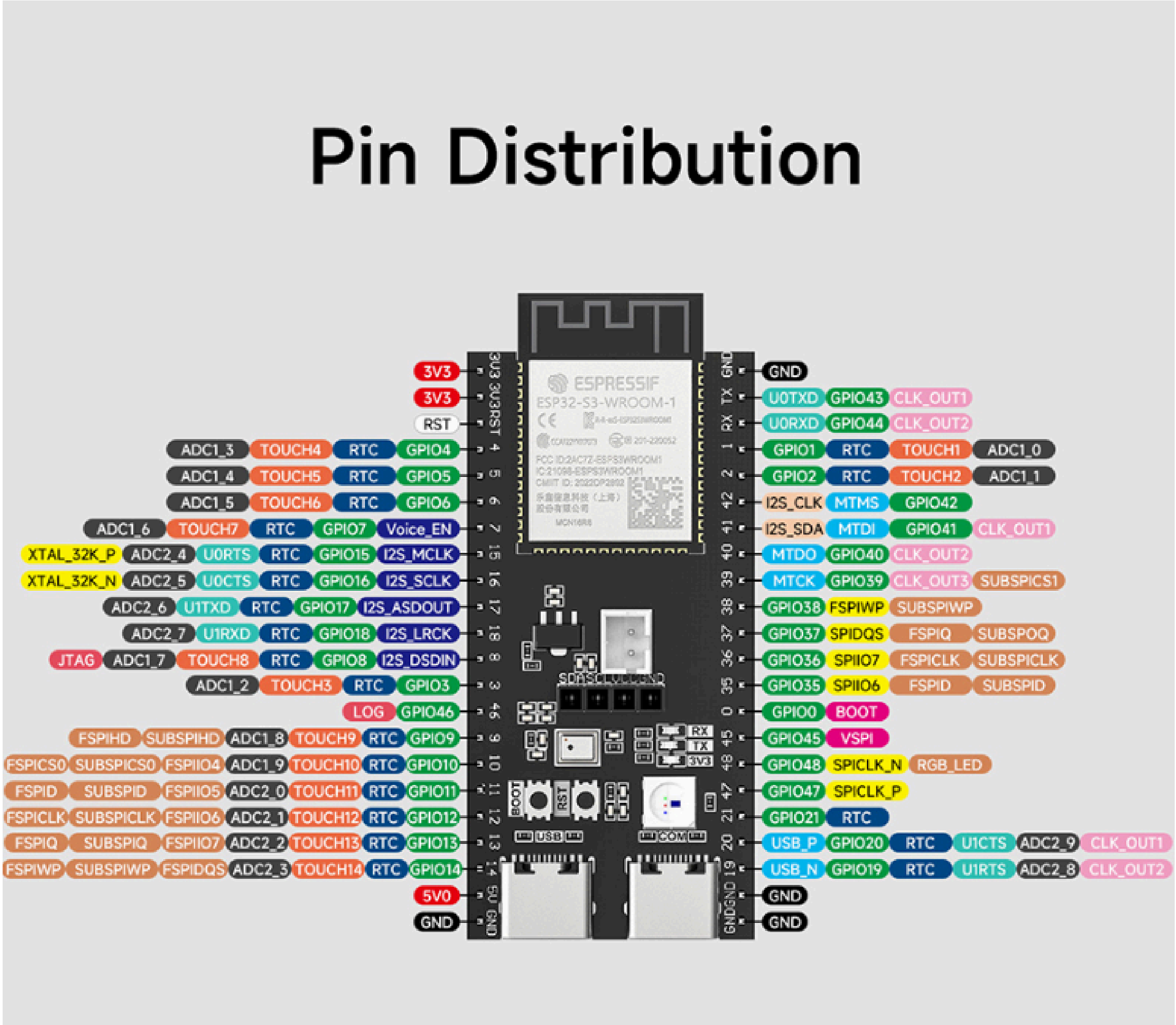
```
┌──────────────────────────────────────────┐
│         ESP32-S3-WROOM-1 module          │
│                                          │
│   ┌──────────────┐    ┌──────────────┐   │
│   │  ESP32-S3    │    │ Octal Flash  │   │
│   │  QFN56 chip  │◄──►│   16 MB      │   │
│   │              │    └──────────────┘   │
│   │              │    ┌──────────────┐   │
│   │              │◄──►│ Octal PSRAM  │   │
│   │              │    │   8 MB       │   │
│   └──────┬───────┘    └──────────────┘   │
│          │                               │
│   ┌──────┴──────────────────────────┐    │
│   │ RF matching network + PCB antenna│   │
│   └─────────────────────────────────┘    │
│                                          │
│   Dimensions: 18.0 × 25.5 × 3.1 mm       │
│   Operating temp: -40°C ~ +85°C          │
└──────────────────────────────────────────┘
```

> Flash and PSRAM are connected to the ESP32-S3 through independent Octal SPI buses. PSRAM occupies GPIO33 ~ GPIO37 (internal connections), and **these 5 GPIOs cannot be used for external wiring**.

3. Complete Pin Mapping

3.1 Fixed-Function Pins (Cannot Be Repurposed)

GPIO	Function	Description
0	BOOT button	Internal pull-up; pressed = 0; low at power-up enters download mode
3	JTAG TMS	USB Serial/JTAG controller
7	NS4150 PA enable	High level = amplifier on
8	I2S DOUT	ES8311 DAC data input (playback path)
15	I2S MCLK	ES8311 master clock (4.096 MHz @ 16 kHz sample rate)
16	I2S BCLK	I2S bit clock
17	I2S DIN	ES8311 ADC data output (recording path)
18	I2S WS	I2S word select (left/right channel switching)
19	USB OTG D-	Native USB OTG data line D-
20	USB OTG D+	Native USB OTG data line D+
33 ~ 37	PSRAM (Octal)	Internally connected to PSRAM in the module, **not usable externally**
41	I2C SDA	Shared I2C data line for ES8311 + SSD1306
42	I2C SCL	Shared I2C clock line for ES8311 + SSD1306
43	UART0 TX	Connected to CH340K RX, used for serial print output
44	UART0 RX	Connected to CH340K TX, used for serial input reception
48	WS2812 RGB LED	Single-wire data pin (RMT driven)



> **Conflict warning**: GPIO43/44 double as UART0 TX/RX. When using `printf()` / `ESP_LOGI()`, these two pins cannot be used for other peripherals at the same time (PWM on GPIO43 will corrupt the serial output).

3.2 Available GPIOs (Allocate as Needed)

The following GPIOs are brought out to the pin headers and do not conflict with on-board peripherals, so they can be freely used in experiments:

GPIO	Special function (optional)	Notes
1	ADC1_CH0	Can be used as ADC / normal IO
2	ADC1_CH1	Can be used as ADC / normal IO
4	ADC1_CH3	Can be used as ADC / normal IO
5	ADC1_CH4	Can be used as ADC / normal IO
6	ADC1_CH5	Can be used as ADC / normal IO

GPIO	Special function (optional)	Notes
9	—	Normal IO
10	FSPICS0	Can be used as normal IO / SPI CS
11	FSPID	SPI MOSI / normal IO
12	FSPICLK	SPI CLK / normal IO
13	FSPIQ	SPI MISO / normal IO
14	—	Normal IO
21	—	Normal IO
38	—	Normal IO
39	—	Normal IO
40	—	Normal IO
45	—	Normal IO (Strapping pin: internal pull-down at power-up)
46	—	Normal IO (Strapping pin: internal pull-down at power-up)
47	—	Normal IO

> **Strapping pin reminder**: GPIO0, GPIO45, and GPIO46 latch their levels at chip power-up to determine the boot mode. GPIO0 is connected to the BOOT button; GPIO45/46 have internal pull-downs. If external pull-up resistors are added, evaluate whether they affect the boot process.

3.3 I2C Device Address Table

ES8311 and SSD1306 share the same I2C bus formed by GPIO41/42:

Device	I2C address	Description
ES8311 audio codec	`0x18`	7-bit address
SSD1306 OLED display	`0x3C`	128×32 monochrome

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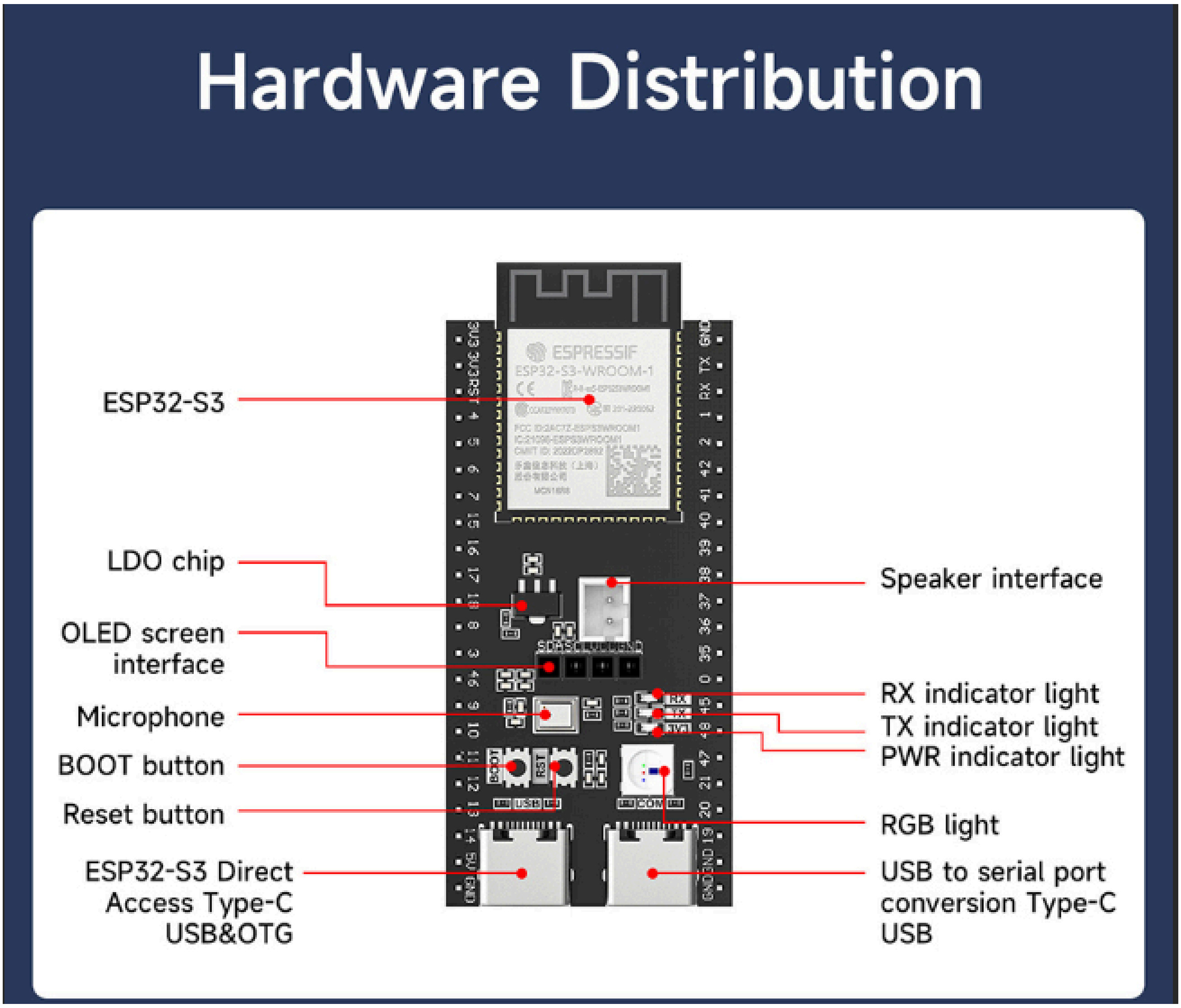
4. Peripheral Details

4.1 Hardware Distribution Diagram



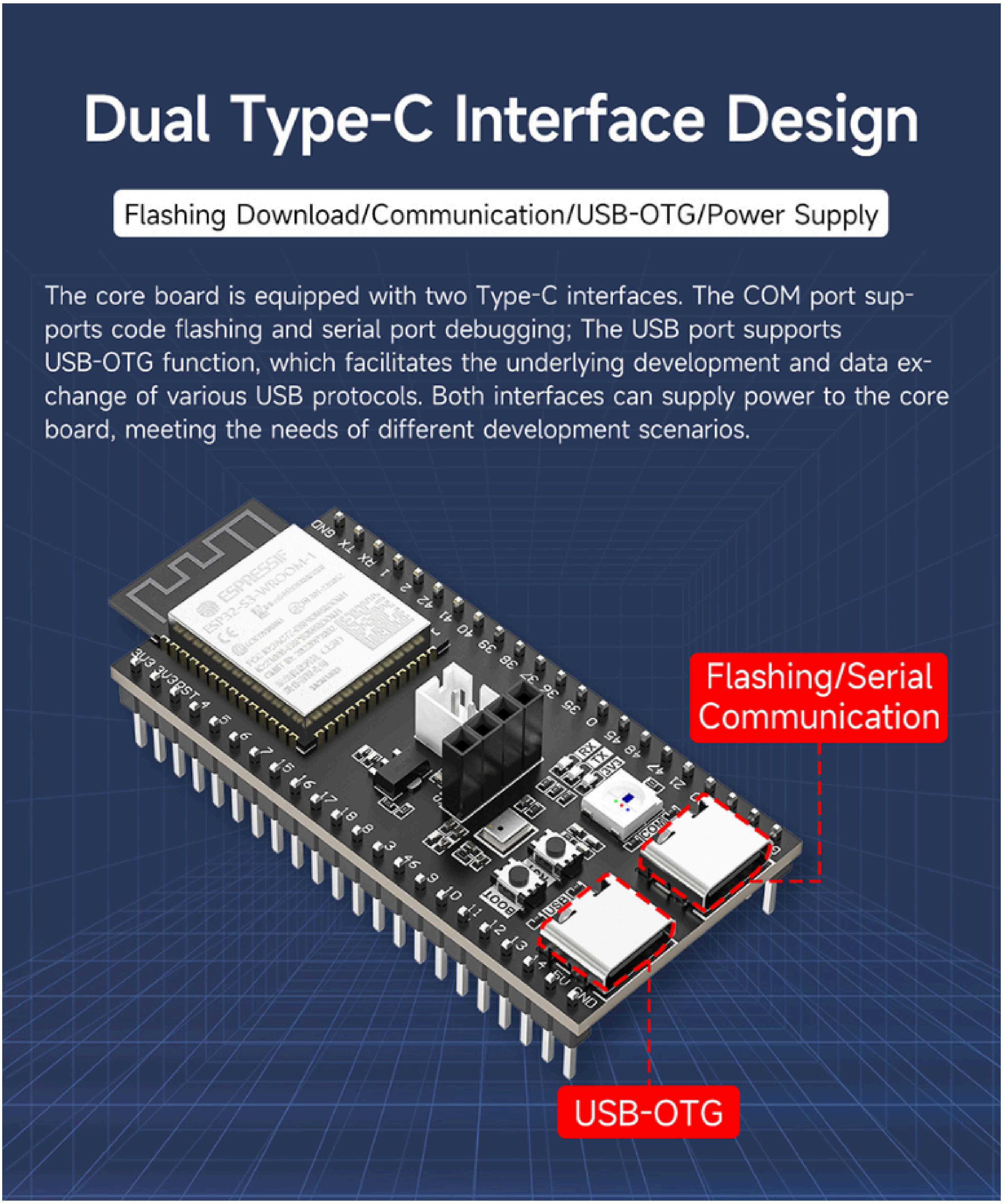
4.2 Power System

```
USB-C (5V) ──→ resettable fuse ──→ Schottky diode ──→ 5V bus ──→ 3.3V LDO ──→ ESP32-S3 + peripherals
                                                      │
                                                      └──→ NS4150 amplifier (direct 5V)
```

Power parameter	Specification
Input voltage	USB-C 5V (typical)
LDO output	3.3V / approx. 800 mA
Amplifier power	Direct 5V (NS4150 supports up to 5.5V)
Resettable fuse	500 mA (auto-disconnects on overcurrent, recovers when cooled)
Reverse polarity protection	Schottky diode (blocks reverse current)

4.3 USB Interfaces

The board has **two USB-C receptacles** with independent functions:



USB port	Function	Connection
USB-UART	Serial debugging + flashing + 5V power	CH340K → ESP32-S3 UART0 (GPIO43/44), with RTS/DTR auto-download circuit

USB port	Function	Connection
USB OTG	Native USB OTG (Device / Host)	ESP32-S3 built-in USB Serial/JTAG controller (GPIO19 = D-, GPIO20 = D+), 3.3V power

> The **USB OTG port** connects directly to the ESP32-S3's built-in USB controller and does not supply power. This port can be used for USB applications such as USB microphone, USB MIDI, USB serial passthrough, and USB drive read/write. It is also the JTAG debug interface; hardware breakpoint debugging can be performed via `idf.py openocd`.

4.4 CH340K USB-to-Serial

Parameter	Description
Chip model	CH340K (by WCH)
Connection	USB-C ↔ UART0 (GPIO43=TX, GPIO44=RX)
Baud rate	Up to 2 Mbps (115200 / 921600 recommended for stability)
Auto-download	Supported (RTS → EN, DTR → BOOT, automatically triggers download mode via transistor circuit)
Drivers	Windows needs the CH340 driver; Linux kernel ≥ 3.x is driver-free; macOS needs a driver

> CH340K is a small-package variant of the CH340 series (10×10 mm), functionally identical to CH340G/C.

4.5 WS2812 RGB LED

Parameter	Description
Model	WS2812-2020 (or compatible SK6812)
Quantity	1 (on-board LED)
Data pin	GPIO48
Power	3.3V (from the LDO, not direct 5V)
Driving method	RMT peripheral (`led_strip` component) or bit-banged GPIO
Color format	GRB (green-red-blue), 8-bit each, 24-bit total
Reset code	≥ 50 μs low level (T0H/T1H timing generated by the RMT hardware)

> The standard WS2812 power supply is 5V, but on this board it still lights up normally at 3.3V (slightly dimmer). For full brightness, use an external LED strip with its own 5V supply.

4.6 Audio System

A complete audio capture and playback chain centered on the ES8311:

```
┌─────────────── Recording path (MIC → ESP32) ───────────────┐
│                                                             │
│  MEMS mic ──→ ES8311 ADC ──→ I2S DIN (GPIO17) ──→ ESP32    │
│                                                             │
└─────────────────────────────────────────────────────────────┘

┌───────────── Playback path (ESP32 → SPK) ──────────────────────┐
│                                                                  │
│  ESP32 ──→ I2S DOUT (GPIO8) ──→ ES8311 DAC ──→ NS4150 PA ──→ SPK │
│                                            (GPIO7 enable)        │
└──────────────────────────────────────────────────────────────────┘

┌─────────────────── Control path (I2C) ───────────────────────┐
│                                                                │
│  ESP32 I2C (GPIO41/42) ──→ ES8311 (0x18)   register config     │
│                        ──→ SSD1306 (0x3C)  display content     │
│                                                                │
└────────────────────────────────────────────────────────────────┘
```

ES8311 Audio Codec

Parameter	Specification
Model	ES8311 (Everest)
Resolution	24-bit ADC / DAC
Sample rate	8 kHz ~ 96 kHz (16 kHz commonly used in this tutorial)
Interface	I2S (Philips standard format, 16-bit) + I2C control
Operating mode	Slave mode (MCLK / BCLK / WS provided by ESP32)
Analog input	MEMS microphone (single-ended / differential)
Analog output	Headphone / Line-out (connected to NS4150 amplifier)
MCLK	4.096 MHz = 256 × 16 kHz (sample rate × 256)
I2C address	`0x18` (7-bit)

NS4150 Amplifier

Parameter	Specification
Model	NS4150 (Class D audio amplifier)
Output power	3W @ 5V / 4Ω (typical)

Parameter	Specification
Enable pin	GPIO7 (high level to enable)
Power supply	Direct 5V (not through the LDO)
Speaker connector	2-pin 1.25 mm PH socket

> Power-on/off timing note: keep the PA off (GPIO7=0) before initializing the ES8311, then pull it high after initialization to avoid pop noise.

4.7 SSD1306 OLED Display

Parameter	Specification
Driver IC	SSD1306
Resolution	128 × 32 pixels (monochrome white)
Interface	I2C (SDA=GPIO41, SCL=GPIO42)
I2C address	`0x3C` (7-bit)
Size	Approx. 0.91 inch
Power	3.3V

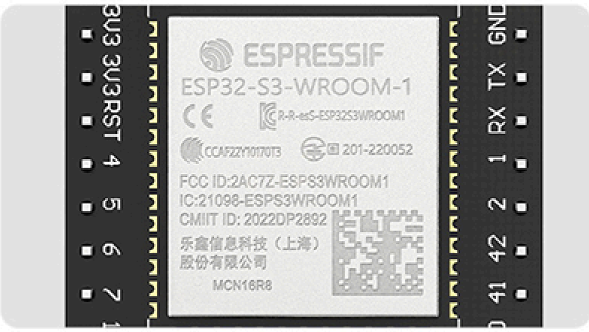
> SSD1306 and ES8311 share the same I2C bus (GPIO41/42). During initialization, create one `i2c_bus_` handle and pass it to both.

4.8 Buttons

Button	GPIO	Default state	Pressed level	Function
BOOT	GPIO0	Pulled high	Low (0)	At runtime = user button; low at power-up = enter download mode
RST	EN (CHIP_PU)	Pulled high	Low (0)	Hardware reset (bypasses the program, directly pulls the chip EN pin low)

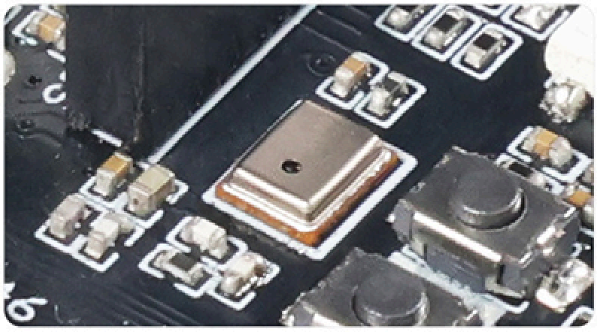
> Manual download mode: **hold BOOT → tap RST → release BOOT**. The auto-download circuit triggers this sequence automatically via RTS/DTR signals, so manual operation is generally unnecessary.

Hardware Features



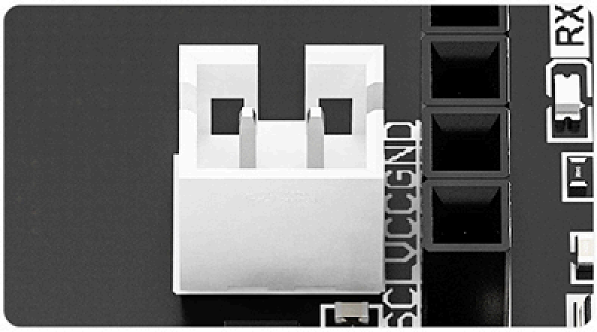
1 ESP32 high-performance chip

Adopting ESP32 high-performance chip, low-power dual core 32-bit CPU, 240MHz main frequency, and computing power up to 600MIPS.



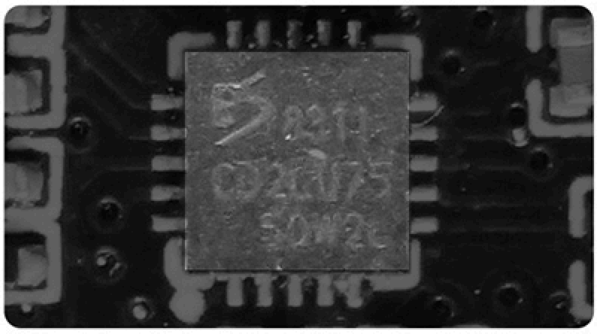
2 High performance microphone

Onboard high-performance microphone, supports clear voice input, seamlessly integrates with AI voice, and achieves intelligent voice interaction.



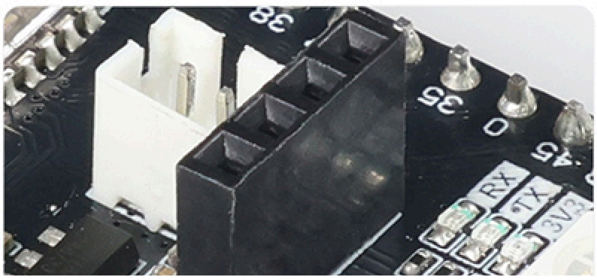
3 Speaker interface

Reserved speaker interface, paired with microphone to unlock recording playback, music playback, and AI voice broadcasting, creating a complete audio loop.



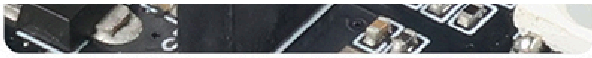
4 Audio decoding chip

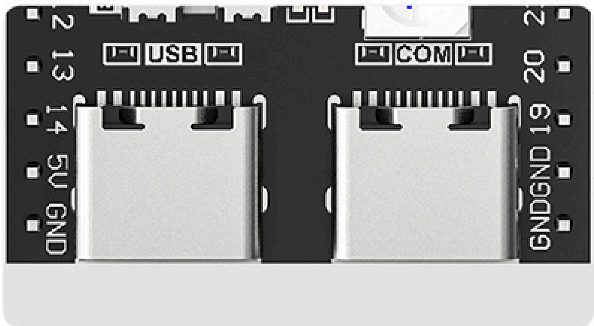
Supports audio capture and playback, unleashes main control computing power, enhances sound quality, and easily expands multimedia functions such as music playback and voice prompts.



5 IIC communication interface

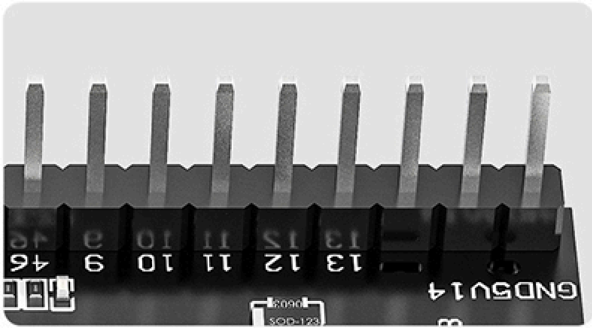
Reserve standard IIC communication interface for easy and quick mounting of various types of sensors and OLED screens.





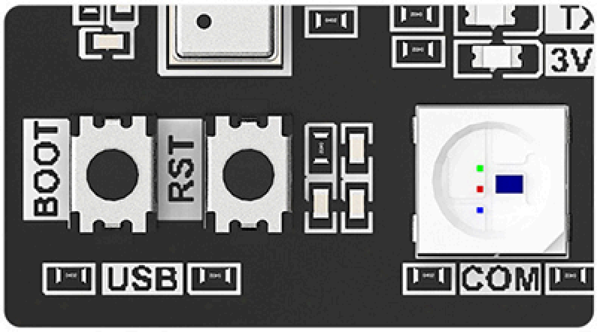
6 Dual Type-C interface

One channel can connect to a computer for one click flashing and serial port debugging; Another channel is specifically designed for USB-OTG.



7 Onboard External Pin Arrangement

Lead out GPIO resources to meet various custom hardware expansion and deep development needs.



8 Onboard custom buttons and RGB lights

Equipped with custom buttons and programmable RGB colorful lights, function buttons and lighting effects can be freely configured through programming.

5. PSRAM and Flash Usage Notes

5.1 Flash Partition Layout (MicroPython Firmware Example)

The MicroPython firmware has a built-in partition table. After flashing, the remaining space is automatically used as the `/flash` file system. Typical layout:

```
0x000000 ┌──────────────────┐
         │  Bootloader      │  ~ 16 KB
0x008000 ├──────────────────┤
         │  Partition Table │  ~ 4 KB
0x009000 ├──────────────────┤
         │  MicroPython     │  ~ 1.5 MB (Python interpreter + hardware drivers)
         │  firmware        │
         ├──────────────────┤  ← firmware end address ≈ 0x1A0000
         │  LittleFS        │  ~ 14 MB (/flash file system, stores .py scripts)
         │  file system     │
0x1000000└──────────────────┘  ← end of 16 MB Flash
```

5.2 PSRAM Memory Mapping

The 8 MB Octal PSRAM is mapped into the ESP32-S3's external address space. After enabling `CONFIG_SPIRAM`, it can be dynamically allocated in FreeRTOS with the `MALLOC_CAP_SPIRAM` flag. After flashing the MicroPython firmware, PSRAM is automatically available (enabled in the `GENERIC_S3_SPIRAM_OCT` firmware). At the Python level, use `gc.mem_free()` to view remaining memory (SRAM only); PSRAM is managed automatically by the interpreter internally.

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6. Typical Development Wiring

6.1 Basic Debugging Wiring

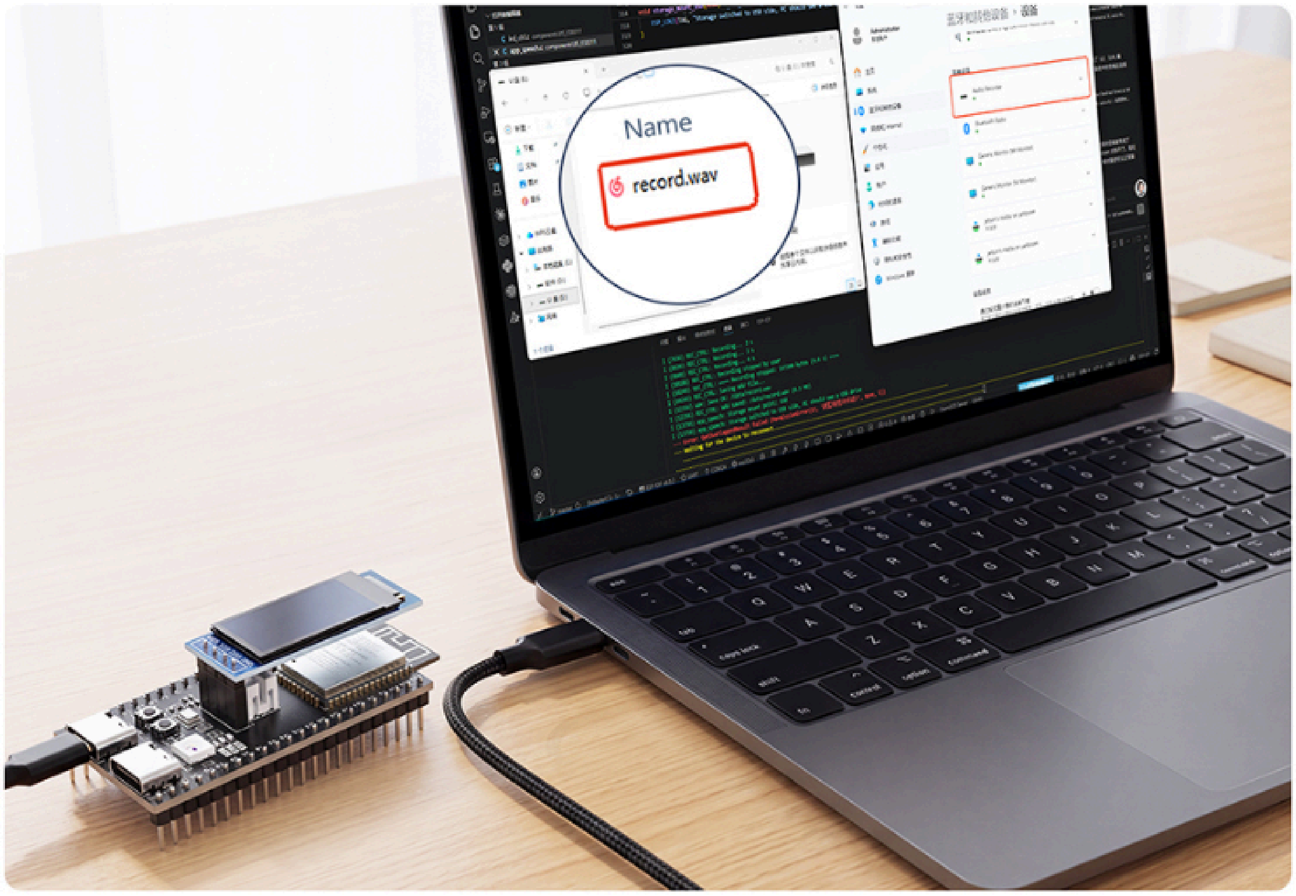
```
ESP32-S3 dev board —— USB-C cable ——→ PC USB port (UART port)
```

No external wiring needed; one USB cable handles power + serial print + flashing.

6.2 Using USB OTG

Simulate USB Storage

The ESP32 core board has a built-in 16MB large capacity Flash. After burning the specified case code, it can be connected to a computer using the USB-OTG interface to simulate a USB flash drive. For example, after recording on the core board, it can be read and played on a PC computer.




```
ESP32-S3 dev board —— USB-C cable ——→ PC USB port (OTG port)
                                        ↑
                        enable USB Device feature in code
```

> The OTG port does not supply power; the board must be powered simultaneously through the UART port (or via 5V/GND on the pin headers).

6.3 Connecting External I2C Devices

The on-board I2C bus already has ES8311 + SSD1306 attached. External devices can be connected in parallel to GPIO41 (SDA) and GPIO42 (SCL) on the pin headers. Note the following:

- The external device's I2C address must not conflict with `0x18` (ES8311) or `0x3C` (SSD1306)
- Pull-up resistors are provided by the on-board devices; no additional pull-ups are needed for external devices
- Bus speed recommended ≤ 400 kHz (Fast Mode)
